

Foundations Of Mathematics

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Contents

1	Sets and Relations	1
2	The Peano Axioms	3
3	Order Relations	5
4	The Axiom Of Choice	6
5	Finite and Infinite Sets	7
6	Well-Ordered Sets	8
7	Quotient Sets and the Integers	9
8	The Real Numbers	11
9	Ordinal Numbers	15
10	Cardinal Numbers	16

Typesetter's Note: Theorems and problems that are denoted by a number preceded by the letter A refer to material from the author's analysis sequence.

1 Sets and Relations

The word *set* is assumed to be a primitive dictionary word. One synonym for this word is *collection*. We hypothesize the existence of a set $\emptyset$ which contains no members, called the *empty set*.

The statement that x is an element of a set A (or x is a member of A ; or x belongs to A) is written $x \in A$.

To designate that B is the set of all elements satisfying property p , we write: $B = \{x : x \text{ has property } p\}$.

To state that a set A is a subset of a set B (meaning that each member of A is also a member of B), we write $A \subseteq B$.

The statement $A = B$ means $A \subseteq B$ and $B \subseteq A$.

If $A \subseteq B$ and $A \neq B$, then we write $A \subset B$ and say that A is a *proper subset* of B .

If A and B are sets, we define the *union* of A and B :

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

and define the *intersection* of A and B :

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

If A and B have no elements in common, then we say that A and B are *disjoint*, and write $A \cap B = \emptyset$.

If $A \subseteq B$, then the *complement* of A in B is:

$$B \setminus A = \{x : x \in B \text{ and } x \notin A\}$$

1. DeMorgan's Laws: If A and B are subsets of a set X , then

- (1) $X \setminus (A \cup B) = (X \setminus A) \cap (X \setminus B)$; and
- (2) $X \setminus (A \cap B) = (X \setminus A) \cup (X \setminus B)$.

2. Theorem: Let A and B be subsets of a set X . These are equivalent:

- (1) $A \subseteq B$;
- (2) $A \cap B = A$;
- (3) $A \cup B = B$;
- (4) $X \setminus B \subseteq X \setminus A$;
- (5) $A \cap (X \setminus B) = \emptyset$; and
- (6) $B \cup (X \setminus A) = X$.

If A and B are sets, then the *product set* $A \times B$ consists of all ordered pairs (a, b) such that $a \in A$ and $b \in B$, i.e.

$$A \times B = \{(a, b) : a \in A, b \in B\}$$

A *relation* on a set A is a subset of $A \times A$.

The *diagonal relation* on a set A is defined:

$$\Delta_A = \{(a, a) : a \in A\}$$

If R is a relation on a set A , then the *converse* of R is defined:

$$R^{-1} = \{(x, y) : (y, x) \in R\}$$

If R and Q are relations on a set A , then the *composition* of R and Q is defined:

$$R \circ Q = \{(x, y) \in A \times A : \text{exists } z \in A \text{ such that } (x, z) \in R \text{ and } (z, y) \in Q\}$$

Let A be a set, and let R be a relation on A . We say that:

- R is *reflexive* if $(a, a) \in R$ for each $a \in A$;
- R is *symmetric* if whenever $(a, b) \in R$, then $(b, a) \in R$, for $a, b \in A$;
- R is *antisymmetric* if whenever both $(a, b) \in R$ and $(b, a) \in R$, then $a = b$, for $a, b \in A$; and
- R is *transitive* if whenever $(a, b) \in R$ and $(b, c) \in R$, then $(a, c) \in R$, for $a, b, c \in A$.

3. Theorem: Let A be a set, and R a relation on A . Then

- (1) R is reflexive if and only if $\Delta_A \subseteq R$;
- (2) R is symmetric if and only if $R = R^{-1}$;
- (3) R is antisymmetric if and only if $R \cap R^{-1} \subseteq \Delta_A$; and
- (4) R is transitive if and only if $R \circ R \subseteq R$.

If $\mathcal{A}$ is a collection of sets, then the *union* of $\mathcal{A}$ is defined $\bigcup\{A : A \in \mathcal{A}\} = \{x : x \in A \text{ for some } A \in \mathcal{A}\}$, and the *intersection* of $\mathcal{A}$ is defined $\bigcap\{A : A \in \mathcal{A}\} = \{x : x \in A \text{ for all } A \in \mathcal{A}\}$.

A collection of sets $\mathcal{A}$ is said to be *indexed* by A provided that for each $\alpha \in A$, there is exactly one $X_\alpha \in \mathcal{A}$ such that $\mathcal{A} = \{X_\alpha : \alpha \in A\}$. The union is denoted $\bigcup_{\alpha \in A} X_\alpha$, and the intersection is denoted $\bigcap_{\alpha \in A} X_\alpha$.

As we will see (using the Axiom of Choice), each collection of sets is an indexed collection.

4. DeMorgan's Laws: Let $\{X_\alpha : \alpha \in A\}$ be a collection of subsets of a set X . Then

$$(1) \quad X \setminus \bigcup_{\alpha \in A} X_\alpha = \bigcap_{\alpha \in A} (X \setminus X_\alpha); \text{ and}$$

$$(2) \quad X \setminus \bigcap_{\alpha \in A} X_\alpha = \bigcup_{\alpha \in A} (X \setminus X_\alpha).$$

A *partition* of a set X is a collection $\mathcal{A}$ of subsets of X such that $X = \bigcup\{A : A \in \mathcal{A}\}$, and if A and B are distinct members of $\mathcal{A}$, then $A \cap B = \emptyset$.

A relation R on a set X is called an *equivalence relation* provided R is reflexive, symmetric, and transitive. The R -classes of X are called *equivalence classes*.

5. Theorem: The classes of an equivalence relation on a set X form a partition of X . Moreover, the elements of a partition of a set X form the equivalence classes of a unique equivalence relation on X .

Let $\mathbb{N} = \{1, 2, 3, \dots\}$ denote the set of all natural numbers.

2 The Peano Axioms

For each $x \in \mathbb{N}$, there is a unique element $x' \in \mathbb{N}$ (called the *successor* of x), such that:

- (a) If $x' = y'$, then $x = y$;
- (b) There is an element of $\mathbb{N}$, which is denoted by 1, such that for each $x \in \mathbb{N}$, $x' \neq 1$; and
- (c) If M is a set which contains 1 and which contains x' whenever it contains x , then M contains every element of $\mathbb{N}$.

Axiom (c) is called **The Axiom of Finite Induction**.

6. Theorem: If $x \in \mathbb{N}$ and $x \neq 1$, then there exists a unique $y \in \mathbb{N}$ such that $x = y'$.

A *function* is a triple (A, B, f) , where A and B are sets, and f is a rule which assigns to each element x of A a unique element $f(x)$ of B . The notation $f : A \rightarrow B$ is frequently used to denote a function (A, B, f) , and if the sets A and B are understood, then f is used to denote the function.

The set A is called the *domain* of the function, and B is called the *codomain* of the function. For $x \in A$, we call $f(x)$ the *image* or *value* of x under the function f . The set of all images of elements of A is a subset of B called the *range* of f .

If $x \neq y$ in A implies that $f(x) \neq f(y)$, then f is called a *one-to-one* function. If the codomain B is also the range of f , then f is said to be an *onto* function.

A function which is both one-to-one and onto is called a *bijection* of A onto B .

The function $1_A : A \rightarrow A$ defined by $1_A(x) = x$ for each $x \in A$, is called the *identity function* on A .

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are functions, then the composition $g \circ f : A \rightarrow C$ is defined by $g \circ f(x) = g(f(x))$ for each $x \in A$.

7. Theorem: Let $f : A \rightarrow B$ be a bijection. Then there is a unique function $g : B \rightarrow A$ such that $g \circ f = 1_A$ and $f \circ g = 1_B$.

The function g in A1.8 is called the *inverse* of f , and is denoted f^{-1} . Notice that it is a bijection from B to A .

If $f : A \rightarrow B$ is a function and $P \subseteq A$, then $f[P] = \{f(x) : x \in P\}$, and for $Q \subseteq B$, then $f^{-1}[Q] = \{x \in A : f(x) \in Q\}$.

8. Theorem: Let $f : X \rightarrow Y$ be a function, and let A and B be subsets of X . Then:

- (1) $f[A \cup B] = f[A] \cup f[B]$;
- (2) $f[A \cap B] \subseteq f[A] \cap f[B]$;
- (3) $f[A \setminus B] \subseteq f[A] \setminus f[B]$; and
- (4) if $A \subseteq B$, then $f[A] \subseteq f[B]$.

9. Theorem: Let $f : X \rightarrow Y$ be a function, and let A and B be subsets of Y . Then:

- (1) $f^{-1}[A \cup B] = f^{-1}[A] \cup f^{-1}[B]$;
- (2) $f^{-1}[A \cap B] = f^{-1}[A] \cap f^{-1}[B]$;
- (3) $f^{-1}[A \setminus B] = f^{-1}[A] \setminus f^{-1}[B]$; and
- (4) If $A \subseteq B$, then $f^{-1}[A] \subseteq f^{-1}[B]$.

10. Theorem: Let $f : X \rightarrow Y$ be a function, and let $\{X_\alpha : \alpha \in A\}$ be a collection of subsets of X . Then

- (1) $f[\bigcup_{\alpha \in A} X_\alpha] = \bigcup_{\alpha \in A} f[X_\alpha]$; and
- (2) $f[\bigcap_{\alpha \in A} X_\alpha] \subseteq \bigcap_{\alpha \in A} f[X_\alpha]$.

11. Theorem: Let $f : X \rightarrow Y$ be a function, and let $\{Y_\beta : \beta \in B\}$ be a collection of subsets of Y . Then:

- (1) $f^{-1}[\bigcup_{\beta \in B} Y_\beta] = \bigcup_{\beta \in B} f^{-1}[Y_\beta]$; and
- (2) $f^{-1}[\bigcap_{\beta \in B} Y_\beta] = \bigcap_{\beta \in B} f^{-1}[Y_\beta]$.

12. Theorem: Let $f : X \rightarrow Y$ be a function, and let $B \subseteq Y$. Then $f^{-1}[Y \setminus B] = X \setminus f^{-1}[B]$.

13. Theorem: There is a unique function $\theta : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that:

- (i) $\theta(x, 1) = x'$ for each $x \in \mathbb{N}$; and
- (ii) $\theta(x, y') = \theta(x, y)'$ for each $x, y \in \mathbb{N}$.

The function θ in A1.15 is called *addition* on $\mathbb{N}$. For $a, b \in \mathbb{N}$, we write $a + b = \theta(a, b)$.

14. Theorem: If $a, b, c \in \mathbb{N}$, then:

- (1) $(a + b) + c = a + (b + c)$ (associative); and
- (2) $a + b = b + a$ (commutative).

15. Theorem: If $a, b, c \in \mathbb{N}$ such that $a + c = b + c$, then $a = b$ (cancellation).

3 Order Relations

A relation $\leq$ on a set X is called a *quasi-order* (we write $x \leq y$ in place of $(x, y) \in \leq$), provided it is reflexive and transitive. If $\leq$ is a quasi-order on a set X , then $(X, \leq)$ is called a *quasi-ordered set*.

A quasi-order on a set X is called a *partial order*, provided it is anti-symmetric. In this case, $(X, \leq)$ is called a *partially ordered set*.

A partial order $\leq$ on a set X is called a *total order*, provided that for $x, y \in X$, either $x \leq y$ or $y \leq x$. In this case, $(X, \leq)$ is called a *totally ordered set*.

If $(X, \leq)$ is a quasi-ordered set, an element $m \in X$ is called a *maximal element* of X , provided $m \leq x$ for $x \in X$ implies $m = x$.

If $(X, \leq)$ is a quasi-ordered set and $A \subseteq X$, then $u \in X$ is called an *upper bound* for A , provided $a \leq u$ for all $a \in A$.

If $(X, \leq)$ is a quasi-ordered set and $C \subseteq X$, then C is called a *chain* in X , provided $(C \times C) \cap \leq$ is a total order on C .

A chain C in a quasi-ordered set $(X, \leq)$ is called a *maximal chain*, provided C is not properly contained in another chain in X ; i.e., C is maximal with respect to $\subseteq$ in the family of chains in X .

If $(X, \leq)$ is a quasi-ordered set and $A \subseteq X$, an element $a \in A$ is called a *first element* of A , provided $a \leq x$ for all $x \in A$.

A quasi-order on a set X is called a *well-order*, provided that each non-empty subset of X has a unique first element. In this case, $(X, \leq)$ is called a *well-ordered set*.

16. Theorem: Each well-ordered set is totally ordered.

Define $\leq$ on $\mathbb{N}$ by $a \leq b$ provided there exists $c \in \mathbb{N}$ such that $b = a + c$ or $b = a$.

17. Theorem: The set $(\mathbb{N}, \leq)$ is a well-ordered set.

18. Recursion Theorem: Let $\mathcal{A}$ be a collection, $e \in \mathcal{A}$, and suppose that for each $n \in \mathbb{N}$, there exists $R_n : \mathcal{A} \rightarrow \mathcal{A}$. Then there exists $F : \mathbb{N} \rightarrow \mathcal{A}$ such that $F(1) = e$ and $F(n+1) = R_{n+1}[F(n)]$ for each $n \in \mathbb{N}$.

19. Example: Let X be a set, and let $f : X \rightarrow X$ be a function. We will use the Recursion Theorem to establish the existence of the n -fold composition of f with itself.

Let $\mathcal{A}$ be the collection of all functions from X into X , and define $R_n : \mathcal{A} \rightarrow \mathcal{A}$ by $R_n(g) = f \circ g$ for each $n \in \mathbb{N}$. Then, by the Recursion Theorem, there exists $F : \mathbb{N} \rightarrow \mathcal{A}$ such that $F(1) = f$ and $F(n+1) = R_{n+1}[F(n)] = f \circ F(n)$ for each $n \in \mathbb{N}$. Define $f^{(n)} = F(n)$ for each $n \in \mathbb{N}$.

4 The Axiom Of Choice

The *Axiom of Choice* states: If $\mathcal{A}$ is a collection of non-empty sets, then there exists a function

$$\phi : \mathcal{A} \rightarrow \bigcup \{A : A \in \mathcal{A}\}$$

such that $\phi(A) \in A$ for each $A \in \mathcal{A}$. The function ϕ is called a *choice function* for the collection $\mathcal{A}$.

The *Cartesian product* of a collection $\mathcal{A}$ of sets is the collection of all choice functions for $\mathcal{A}$. It is denoted $\prod \{A : A \in \mathcal{A}\}$. For each ϕ in this product and each $A \in \mathcal{A}$, the value $\phi(A)$ is called the *A coordinate* of ϕ . For each $B \in \mathcal{A}$, the function $\pi_B : \prod \{A : A \in \mathcal{A}\} \rightarrow B$ defined by $\pi_B(\phi) = \phi(B)$ is called the *B projection*.

If $(X, \leq)$ is a partially ordered set and $A \subseteq X$, then an element $b \in X$ is called a *least upper bound* for A , provided b is an upper bound for A and $b \leq u$ for each upper bound u of A . Note that if a least upper bound exists for a subset of a partially ordered set, then it is unique.

If $(X, \leq)$ is a non-empty partially ordered set such that each totally ordered subset of X has a least upper bound $b \in X$, and if $f : X \rightarrow X$ is a function such that $x \leq f(x)$ for all $x \in X$, then a subset A of X is said to be *admissible* (with respect to b), provided:

- (a) $b \in A$;
- (b) $f[A] \subseteq A$; and
- (c) every least upper bound of a totally ordered subset of A belongs to A .

20. Lemma: Let $(X, \leq)$ be a non-empty partially ordered set such that each totally ordered subset of X has a least upper bound, let $f : X \rightarrow X$ be a function such that $x \leq f(x)$ for each $x \in X$, let $b \in X$, and let $\mathcal{A}$ be the family of all admissible subsets of X (with respect to b). Then:

- (a) $\mathcal{A} \neq \emptyset$;

- (b) $(\mathcal{A}, \subseteq)$ is a partially ordered set with $M = \bigcap \{A : A \in \mathcal{A}\}$ as a unique minimal element;
 - (c) if $x \in M$, then $b \leq x$;
 - (d) Let $P = \{x \in M : \text{if } y \in M \text{ and } y < x, \text{ then } f(y) \leq x\}$. Now if $x \in P$ and $z \in M$, then either $z \leq x$ or $f(x) \leq z$;
 - (e) P is an admissible subset of M , and hence $P = M$; and
 - (f) M is totally ordered.
- 21. Lemma:** Let $(X, \leq)$ be a non-empty partially ordered set such that each totally ordered subset of X has a least upper bound. If $f : X \rightarrow X$ is such that $x \leq f(x)$ for all $x \in X$, then there exists $p \in X$ such that $f(p) = p$.
- 22. Theorem:** Each of the following is equivalent to the Axiom of Choice:
- (a) **Kuratowski's Lemma:** Each chain in a partially ordered set is contained in a maximal chain.
 - (b) **Hausdorff Maximal Principle:** Each partially ordered set contains a maximal chain.
 - (c) **Zorn's Lemma:** If each chain in a partially ordered set $(X, \leq)$ has an upper bound, then X has a maximal member.
 - (d) **The Well-Ordering Principle:** Each set can be well-ordered.

5 Finite and Infinite Sets

A set X is said to be *infinite* provided there exists a proper subset $B \subset X$ and a one-to-one function $f : X \rightarrow B$ from X onto B . Otherwise, we say that X is *finite*.

For $n \in \mathbb{N}$, we use $\downarrow n$ to denote the set $\{m \in \mathbb{N} : m \leq n\}$.

- 23. Lemma:** Let X be a set. Then exactly one of the following holds:
- (i) There exists a one-to-one function $\phi : \downarrow n \rightarrow X$ of $\downarrow n$ onto X ; or
 - (ii) There exists a one-to-one function $\psi : \mathbb{N} \rightarrow X$ of $\mathbb{N}$ into X .
- 24. Theorem:** Let X be a non-empty set. Then X is finite if and only if there exists $n \in \mathbb{N}$ and a one-to-one onto function $f : X \rightarrow \downarrow n$.

A set X is said to be *denumerable* if there exists a one-to-one function $\phi : X \rightarrow \mathbb{N}$ from X onto $\mathbb{N}$.

A set X is said to be *countable* if it is either finite or denumerable. It is *uncountable* otherwise.

- 25. Theorem:** A set X is infinite if and only if X contains a denumerable subset.

- 26. Theorem:** If the domain of a function is countable, then the range is also countable.
- 27. Theorem:** If $\mathcal{A}$ is a countable family of countable sets, then $\bigcup\{A : A \in \mathcal{A}\}$ is countable.

The following Lemma will be useful in the proof of A4.7:

- 28. Lemma:** Let $\{x_n\}$ be a sequence in $\mathbb{N}$. Then there exists an increasing sequence $\{p_n\}$ such that $x_n < p_n$ for each $n \in \mathbb{N}$.

Proof: We will apply the Recursion Theorem. Let $p_1 = x_1 + 1 = e$, and define $R_n : \mathbb{N} \rightarrow \mathbb{N}$ by $R_n(x) = x + x_n + 1$. Then there exists a function $F : \mathbb{N} \rightarrow \mathbb{N}$ such that $F(1) = p_1$ and $F(n+1) = R_{n+1}[F(n)]$ for each $n \in \mathbb{N}$. Let $p_n = F(n)$ for each $n \in \mathbb{N}$. Then $p_{n+1} = F(n+1) = p_n + x_{n+1} + 1$.

- 29. Theorem:** Let D be a denumerable set, $\mathcal{F}$ the family of all finite subsets of D , and let $\mathcal{S}$ be the family of all subsets of D . Then $\mathcal{F}$ is denumerable and $\mathcal{S}$ is uncountable.

6 Well-Ordered Sets

If $(X, \leq)$ is a well-ordered set and $a \in X$, then $x < a$ means that $x \leq a$ and $x \neq a$. The set $L(a) = \{x \in X : x < a\}$ is called the *initial interval* determined by a .

If $(X, \leq)$ is a well-ordered set and $I \subseteq X$, then I is said to be an *ideal* of X , provided $L(a) \subseteq I$ for each $a \in I$.

If $(X, \leq)$ is a well-ordered set, then X and $\emptyset$ are ideals of X . Observe that $\emptyset$ is an initial interval determined by the first element of X , and X is not an initial interval. Also, $L(a)$ is an ideal for each $a \in X$.

If $(X, \leq)$ is a well-ordered set, then $\mathcal{I}(X)$ denotes the family of all ideals of X , and $\mathcal{J}(X)$ denotes the family of all initial intervals of X .

- 30. Theorem:** Let $(X, \leq)$ be a well-ordered set. Then:

- (a) The intersection of ideals of X is an ideal of X ;
- (b) The union of ideals of X is an ideal of X ; and
- (c) $\mathcal{J}(X) = \mathcal{I}(X) \setminus \{X\}$.

- 31. Theorem:** If $(X, \leq)$ is a well-ordered set, then $(\mathcal{I}(X), \subseteq)$ is a well-ordered set.

If $(X, \leq)$ and $(X', \leq')$ are well-ordered sets, a function $f : X \rightarrow X'$ is called a *monomorphism*, provided f is one-to-one and order-preserving; i.e., $x \leq y$ in X implies that $f(x) \leq' f(y)$ in X' . If f also maps X onto X' , then f is called an *isomorphism*.

- 32. Theorem:** If $(X, \leq)$ is a well-ordered set, then the function $a \mapsto L(a)$ of $(X, \leq)$ onto $(\mathcal{J}(X), \subseteq)$ is an isomorphism.
- 33. Theorem:** Let $(X, \leq)$ be a well-ordered set, and let Σ be a family of ideals of X satisfying:
- (a) Any union of members of Σ is a member of Σ ; and
 - (b) If $L(a) \in \Sigma$, then $L(a) \cup \{a\} \in \Sigma$.
 - (c) Then $\Sigma = \mathcal{I}(X)$ and, in particular, $X \in \Sigma$.
- 34. Lemma:** Let $(X, \leq)$ and $(X', \leq')$ be well-ordered sets, and let $\phi : X \rightarrow X'$ be an isomorphism of X onto an ideal of X' . Then any monomorphism $f : X \rightarrow X'$ satisfies the condition that $\phi(x) \leq' f(x)$ for each $x \in X$. In particular, there can be at most one isomorphism between an ideal of X and an ideal of X' .
- 35. Theorem:** Let $(X, \leq)$ and $(X', \leq')$ be well-ordered sets. Then exactly one of the following holds:
- (a) There is a unique isomorphism of X onto X' ;
 - (b) There is a unique isomorphism of X onto an initial interval of X' ; and
 - (c) There is a unique isomorphism of X' onto an initial interval of X .
- 36. Corollary:** Let $(X, \leq)$ be a well-ordered set, and let $A \subseteq X$. Then A is isomorphic to X or to an initial interval of X .
- 37. Corollary:** Let $(X, \leq)$ be a well-ordered set. Then no initial interval of X is isomorphic to X .
- 38. Corollary:** The class of all well-ordered sets is well-ordered, if we define $X \leq X'$ to mean that X is isomorphic to an ideal of X' (and $X = X'$ means that X is isomorphic to X').
- 39. Transfinite Induction:** Let $(X, \leq)$ be a well-ordered set, and let $A \subseteq X$. If $L(x) \subseteq A$ implies that $x \in A$ for each $x \in X$, then $A = X$.

Note that if $(\mathbb{N}, \leq)$ is the positive integers with its usual order, then A39 is equivalent to what is usually referred to as *strong finite induction*.

7 Quotient Sets and the Integers

40. Theorem: There is a unique function $\phi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$ such that:

- (1) $\phi(x, 1) = x$ for each $x \in \mathbb{N}$; and
- (2) $\phi(x, y + 1) = \phi(x, y) + x$ for each $x, y \in \mathbb{N}$.

The function ϕ in A6.1 is called *multiplication* on $\mathbb{N}$, and we write xy for $\phi(x, y)$ for each $x, y \in \mathbb{N}$.

41. Theorem: If $a, b, c \in \mathbb{N}$, then:

- (1) $(ab)c = a(bc)$ (associative);
- (2) $ab = ba$ (commutative);
- (3) $a(b + c) = ab + ac$ (distributive); and
- (4) If $ac = bc$, then $a = b$ (cancellation).

If X is a set and σ is an equivalence relation on X , then X/σ denotes the set of σ -classes of X . We call X/σ the *quotient set* of X mod σ . Then function $\eta : X \rightarrow X/\sigma$, defined by letting $\eta(x)$ be the σ -class containing x for each $x \in X$, is called the *natural map*. Observe that for $x, y \in X$, $\eta(x) = \eta(y)$ if and only if $(x, y) \in \sigma$. Moreover, for each $x \in X$, $\eta^{-1}\eta(x) = \{y \in X : (x, y) \in \sigma\}$ (the σ -class of x).

Define a relation τ on $\mathbb{N} \times \mathbb{N}$ as follows:

$$\tau = \{((a, b), (x, y)) \in (\mathbb{N} \times \mathbb{N}) \times (\mathbb{N} \times \mathbb{N}) : a + y = x + b\}$$

42. Theorem: The relation τ is an equivalence relation on $\mathbb{N} \times \mathbb{N}$.

Let $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\tau$, and let $\mu : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{Z}$ be the natural map.

43. Theorem: The map $\beta : \mathbb{N} \rightarrow \mathbb{Z}$ defined by $\beta(x) = \mu((x + 1, 1))$ is a one-to-one function.

The set $\mathbb{Z}$ is called the *integers*. In view of A6.4, we will identify $\mathbb{N}$ with its image under β in $\mathbb{Z}$.

Let $\Delta = \{(a, a) : a \in \mathbb{N}\}$.

44. Theorem: The set Δ is a τ -class in $\mathbb{N} \times \mathbb{N}$.

The image $\mu[\Delta]$ in $\mathbb{Z}$ is called *zero* and is denoted by 0.

45. Lemma: If $a, b, x \in \mathbb{N}$, then $\mu(a + x, b + x) = \mu(a, b)$.

We define *addition* on $\mathbb{Z}$ by $\mu(a, b) + \mu(c, d) = \mu(a + c, b + d)$ for $a, b, c, d \in \mathbb{N}$.

46. Theorem: Addition on $\mathbb{Z}$ is well-defined. Moreover, if $x, y, z \in \mathbb{Z}$, then:

- (1) $(x + y) + z = x + (y + z)$ (associative);
- (2) $x + y = y + x$ (commutative);
- (3) $x + z = y + z$ implies $x = y$ (cancellative); and
- (4) if $x, y \in \mathbb{N}$, then $x + y \in \mathbb{N}$.

If $z = \mu(a, b)$ in $\mathbb{Z}$, then we write $-z = \mu(b, a)$. For $x, y \in \mathbb{Z}$, we write $x - y = x + (-y)$.

47. Theorem: The set $\mathbb{Z} = \mathbb{N} \cup \{0\} \cup \{-n : n \in \mathbb{N}\}$, and these sets are disjoint.

48. Theorem: If $x \in \mathbb{Z}$, then

- (1) $x + 0 = x$; and
- (2) $x + (-x) = 0$.

Note: $(\mathbb{Z}, +)$ is an abelian group whose identity is 0.

We define $\leq$ on $\mathbb{Z}$ by $x \leq y$, provided that either $x = y$ or $y - x \in \mathbb{N}$. We say that $z \in \mathbb{Z}$ is *positive* provided $z \in \mathbb{N}$.

49. Theorem: The relation $\leq$ is a total order on $\mathbb{Z}$. Moreover,

- (1) $x \in \mathbb{Z}$ is positive if and only if $0 < x$ ($0 \leq x$ and $x \neq 0$); and
- (2) If $x \leq y$ and $z \in \mathbb{Z}$, then $x + z \leq y + z$.

We define *multiplication* on $\mathbb{Z}$ by

$$\mu(a, b) \cdot \mu(c, d) = \mu(ac + bd, bc + ad)$$

50. Theorem: Multiplication on $\mathbb{Z}$ is well-defined. Moreover, if $x, y, z \in \mathbb{Z}$, then:

- (1) $(xy)z = x(yz)$ (associative);
- (2) $xy = yx$ (commutative);
- (3) If $z \neq 0$ and $xz = yz$, then $x = y$ (cancellative);
- (4) $x \cdot 0 = 0$;
- (5) $x \cdot 1 = x$; and
- (6) $x(y + z) = xy + xz$ (distributive).

51. Theorem: Let $x, y, z \in \mathbb{Z}$.

- (1) If $x \leq y$ and $0 \leq z$, then $xz \leq yz$; and
- (2) If $x \leq y$ and $z \leq 0$, then $yz \leq xz$.

8 The Real Numbers

Let $\mathbb{Z}_* = \mathbb{Z} \setminus \{0\}$, and let $\sigma = \{((a, b), (x, y)) \in (\mathbb{Z} \times \mathbb{Z}_*) \times (\mathbb{Z} \times \mathbb{Z}_*) : ay = bx\}$.

52. Theorem: The relation σ is an equivalence relation on $\mathbb{Z} \times \mathbb{Z}_*$.

Note:

- (1) $\sigma(0, 1) = \sigma(0, x)$ for all $x \in \mathbb{Z}_*$;
- (2) If $\sigma(0, 1) = \sigma(a, b)$, then $a = 0$; and
- (3) If $x \neq 0$, then $\sigma(a, b) = \sigma(xa, xb)$.

The quotient $\mathbb{Q} = (\mathbb{Z} \times \mathbb{Z}_*)/\sigma$ is called the *set of rational numbers*. Let $\sigma[a, b]$ denote the σ -class of $(a, b) \in \mathbb{Z} \times \mathbb{Z}_*$.

Define addition on $\mathbb{Q}$ by $\sigma[a, b] + \sigma[c, d] = \sigma[ad + bc, bd]$; and multiplication by $\sigma[a, b] \cdot \sigma[c, d] = \sigma[ac, bd]$.

53. Theorem: Addition and multiplication are well-defined functions from $\mathbb{Q} \times \mathbb{Q}$ into $\mathbb{Q}$.

54. Theorem: Addition and multiplication on $\mathbb{Q}$ are associative and commutative. Moreover, $1 = \sigma[1, 1]$ is a multiplicative identity and $0 = \sigma[0, 1]$ is an additive identity.

If $x \in \mathbb{Q}$, $x = \sigma[a, b]$, let $-x = \sigma[-a, b]$ ($= \sigma[a, -b]$), and if $x \neq 0$, let $x^{-1} = \sigma[b, a]$.

7.4. Theorem:

- (i) If $x \in \mathbb{Q}$, then $x + (-x) = 0$;
- (ii) If $x \in \mathbb{Q}$, $x \neq 0$, then $xx^{-1} = 1$;
- (iii) If $x, y, z \in \mathbb{Q}$, then $x(y + z) = xy + xz$ (distributive); and
- (iv) If $x \in \mathbb{Q}$, then $x \cdot 0 = 0$.

For $p \in \mathbb{Q}$, define $0 \leq p$ provided either $p = 0$ or $p = \sigma[a, b]$, where $0 < a$ and $0 < b$ in $\mathbb{Z}$. For $x, y \in \mathbb{Q}$, define $x \leq y$ if $0 \leq y - x$ ($= y + (-x)$).

55. Theorem: If $x, y, z \in \mathbb{Q}$ with $x \leq y$, then

- (i) $x + z \leq y + z$;
- (ii) if $0 < z$, then $xz \leq yz$; and
- (iii) if $z < 0$, then $yz \leq xz$.

We say that $x \in \mathbb{Q}$ is *positive* if $0 < x$, and *negative* if $x < 0$.

56. Theorem: The relation $\leq$ is a total order on $\mathbb{Q}$.

57. Theorem: The map $\psi : \mathbb{Z} \rightarrow \mathbb{Q}$ defined by $\psi(x) = \sigma[x, 1]$ is a one-to-one map of $\mathbb{Z}$ into $\mathbb{Q}$ such that $\psi(x + y) = \psi(x) + \psi(y)$ and $\psi(xy) = \psi(x)\psi(y)$ for each $x, y \in \mathbb{Z}$.

We identify $\mathbb{Z}$ with its image under ψ in $\mathbb{Q}$.

For $x \in \mathbb{Q}$, define *absolute value* of x :

$$|x| = \begin{cases} x & \text{if } 0 \leq x; \\ -x & \text{if } x < 0 \end{cases}$$

58. Theorem: If $x, y, z \in \mathbb{Q}$, then $|x + y| \leq |x| + |y|$.

Note: $(\mathbb{Q}, +, \cdot, \leq)$ is an ordered field.

Let $\mathbb{Q}^{\mathbb{N}} = \{f : \mathbb{N} \rightarrow \mathbb{Q}\}$; i.e., the set of all sequences in $\mathbb{Q}$. We will adopt the usual convention of writing $x_n = f(n)$, and say that $\{x_n\}$ is a sequence.

A sequence $\{x_n\}$ is called a *Cauchy sequence*, provided for each $\epsilon \in \mathbb{Q}$ such that $0 < \epsilon$, there exists $p \in \mathbb{N}$ such that $|x_n - x_m| < \epsilon$ when $p < n, m$. We denote by $\mathbb{Q}_{\star}^{\mathbb{N}}$ the set of all Cauchy sequences in $\mathbb{Q}$.

Define $\rho = \{(\{x_n\}, \{y_n\}) \in \mathbb{Q}_{\star}^{\mathbb{N}} \times \mathbb{Q}_{\star}^{\mathbb{N}} : \text{for each } 0 < \epsilon \text{ in } \mathbb{Q}, \text{ there exists } p \in \mathbb{N} \text{ such that } |x_n - y_n| < \epsilon, \text{ when } p < n\}$.

Note: Each Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$ is bounded; i.e., there exists $r \in \mathbb{Q}$ such that $|x_n| \leq r$ for all $n \in \mathbb{N}$.

59. Theorem: The relation ρ is an equivalence relation on $\mathbb{Q}_{\star}^{\mathbb{N}}$.

The quotient $\mathbb{R} = \mathbb{Q}_{\star}^{\mathbb{N}}/\rho$ is called the *set of real numbers*. For a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$, we let $\rho[x_n]$ denote the ρ -class of $\{x_n\}$.

Note that for $r \in \mathbb{Q}$, the sequence $\{x_n\}$ such that $x_n = r$ for all $n \in \mathbb{N}$ is a Cauchy sequence (which we denote by $\{r\}$).

Define *addition* on $\mathbb{R}$ by $\rho[x_n] + \rho[y_n] = \rho[x_n + y_n]$; and define *multiplication* on $\mathbb{R}$ by $\rho[x_n] \cdot \rho[y_n] = \rho[x_n y_n]$.

60. Theorem: Multiplication and addition are well-defined functions $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$.

61. Theorem: Multiplication and addition on $\mathbb{R}$ are associative and commutative. Moreover, $1 = \rho[1]$ is a multiplicative identity and $0 = \rho[0]$ is an additive identity.

62. Lemma: If $x \neq 0$ in $\mathbb{R}$, then there exists a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$ such that $x = \rho[x_n]$ and $x_n \neq 0$ for all $n \in \mathbb{N}$.

If $x \in \mathbb{R}$, $x = \rho[x_n]$, define $-x = \rho[-x_n]$.

If $x \in \mathbb{R}$ and $x \neq 0$, let $\{x_n\}$ be a Cauchy sequence in $\mathbb{Q}$ such that $x_n \neq 0$ for all $n \in \mathbb{N}$, and define $x^{-1} = \rho[x_n^{-1}]$.

63. Theorem: If $x \in \mathbb{R}$, then $-x$ is well-defined. Moreover; if $x \neq 0$, then x^{-1} is well-defined.

64. Theorem: Let $x, y, z \in \mathbb{R}$.

- (i) $x + (-x) = 0$;
- (ii) if $x \neq 0$, then $xx^{-1} = 1$;
- (iii) $x(y + z) = xy + xz$ (distributive); and
- (iv) $x \cdot 0 = 0$.

For $p \in \mathbb{R}$, define $0 \leq p$ provided there exists a Cauchy sequence $\{x_n\}$ in $\mathbb{Q}$ such that $0 \leq x_n$ for all $n \in \mathbb{N}$ and $p = \rho[x_n]$. Define $0 < p$ if $0 \leq p$ and $p \neq 0$. For $x, y \in \mathbb{R}$, define $x \leq y$ provided $0 \leq y - x (= y + (-x))$.

65. Theorem: If $x, y, z \in \mathbb{R}$ with $x \leq y$, then

- (i) $x + z \leq y + z$;
- (ii) if $0 < z$, then $xz \leq yz$; and
- (iii) if $z < 0$, then $yz \leq xz$.

We say that $x \in \mathbb{R}$ is *positive* if $0 < x$, and *negative* if $x < 0$.

66. Theorem: The relation $\leq$ is a total order on $\mathbb{R}$.

67. Theorem: The map $\phi : \mathbb{Q} \rightarrow \mathbb{R}$ defined by $\phi(a) = \rho[a]$ is a one-to-one map of $\mathbb{Q}$ into $\mathbb{R}$ such that $\phi(a + b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a) \cdot \phi(b)$ for each $a, b \in \mathbb{Q}$.

We identify $\mathbb{Q}$ with its image under ϕ in $\mathbb{R}$.

Note: $(\mathbb{R}, +, \cdot, \leq)$ is an ordered field.

For $x \in \mathbb{R}$, define the *absolute value* of x :

$$|x| = \begin{cases} x & \text{if } 0 \leq x; \\ -x & \text{if } x < 0 \end{cases}$$

A68. Theorem: If $x, y \in \mathbb{R}$, then $|x + y| \leq |x| + |y|$.

A69. Theorem: If $x < y$ in $\mathbb{R}$, then there exists $q \in \mathbb{Q}$ such that $x < q < y$.

A sequence $\{x_n\}$ in $\mathbb{R}$ is called a *Cauchy sequence* provided that for each $0 < \epsilon$ in $\mathbb{R}$, there exists $p \in \mathbb{N}$ such that $|x_n - x_m| < \epsilon$ when $p < n, m$.

A sequence $\{x_n\}$ in $\mathbb{R}$ is said to *converge* to $x \in \mathbb{R}$ if for each $0 < \epsilon$ in $\mathbb{R}$, there exists $p \in \mathbb{N}$ such that $|x_n - x| < \epsilon$ when $p < n$.

A70. Archimedean Property: If $r \in \mathbb{R}$, then there exists $n \in \mathbb{N}$ such that $r < n$.

A71. Completeness: Each Cauchy sequence in $\mathbb{R}$ converges.

A72. Theorem: Each subset of $\mathbb{R}$ which has an upper bound has a least upper bound.

A73. Theorem: The set $\mathbb{Q}$ is denumerable.

A74. Theorem: The set $\mathbb{R}$ is non-denumerable.

9 Ordinal Numbers

An *ordinal number* is a set P satisfying:

- (a) If $x \in P$ and $y \in P$, then either $x \in y$ or $y \in x$, or $x = y$; and
- (b) If $x \in y$ and $y \in P$, then $x \in P$.

75. Example: $P = \{\emptyset, \{\emptyset\}, \{\emptyset, \{\emptyset\}\}\}$ is an ordinal number.

Some results from set theory are:

- (a) **Axiom:** If A is a set, then there exists $u \in A$ such that $u \cap A = \emptyset$.
- (b) **Theorem:** If A is a set, then $A \notin A$.
- (c) **Theorem:** If A and B are sets, then either $A \notin B$ or $B \notin A$.

Note that if P is an ordinal number, then $\in$ is not a quasi-order on P , since $x \notin x$ for any $x \in P$.

76. Theorem: Let P be an ordinal number, and let A be a non-empty subset of P . Then there is a unique $a \in A$ such that either $a \in x$ or $a = x$ for each $x \in A$.

The element a in A8.1 is called the *first element* of A .

77. Theorem: If P is an ordinal number, then $\emptyset$ is the first element of P .

78. Theorem: If P is an ordinal number and $x \in P$, then x is an ordinal number.

79. Theorem: If P and Q are distinct ordinal numbers, then $P \subset Q$ if and only if $P \in Q$.

80. Theorem: Let P and Q be distinct ordinal numbers. Then either $P \subset Q$ or $Q \subset P$.

81. Theorem: The class of all ordinal numbers is well-ordered by $\subseteq$.

82. Theorem: If $\mathcal{E}$ is a class of ordinal numbers, then $\mathcal{E}$ is well-ordered by $\subseteq$, and in particular $\mathcal{E}$ has a first element.

83. Theorem: The class of all ordinal numbers is not a set.

Let $\mathcal{O}$ denote the class of all ordinal numbers.

84. Theorem: If P is an ordinal number, then the set $L(P)$ of initial intervals determined by P in $(\mathcal{O}, \subseteq)$ is P . In particular, $L(P)$ is a set.

85. Theorem: If E is a set of ordinal numbers, then E has a least upper bound in $(\mathcal{O}, \subseteq)$.

86. Theorem: If $(X, \leq)$ is a well-ordered set, then $(X, \leq)$ is isomorphic to $(P, \subseteq)$ for some unique ordinal number P .

In A86, we call P the *ordinal* of $(X, \leq)$, and denote $P = \text{ord}(X, \leq)$.

87. Theorem: Let $(X, \leq)$ be a well-ordered set, and suppose $m \notin X$. Then $(X \cup \{m\}, \leq)$ is well-ordered if $\leq$ is extended to $X \cup \{m\}$ so that $x \leq m$ for all $x \in X \cup \{m\}$. Moreover, $\text{ord}(X \cup \{m\}, \leq) = \text{ord}(X, \leq) \cup \{\text{ord}(X, \leq)\}$.

88. Theorem: If $P \in \mathcal{O}$, then $P \cup \{P\}$ is the successor of P in $(\mathcal{O}, \subseteq)$.

An ordinal $P \neq \emptyset$ is called a *limit ordinal* provided P has no predecessor in $(\mathcal{O}, \subseteq)$.

Let $(\mathbb{N}^*, \leq)$ denote the non-negative integers with the usual order. Let $\omega = \text{ord}(\mathbb{N}^*, \leq)$. Let $\bar{n}$ denote $(L(n), \leq)$ for each $n \in \mathbb{N}^*$.

For $n \in \mathbb{N}^*$, we will use $\underline{n}$ to denote $\text{ord } \bar{n}$.

For sets A and B , we use $A \dot{\cup} B$ to denote the disjoint union of A and B .

89. Theorem: Let A and B be ordinal numbers. Define $\leq$ on $A \dot{\cup} B$ as follows: for $x, y \in A \dot{\cup} B$,

$x \leq y$ if either x and y are both in A or both in B and $x \subseteq y$; and

$x < y$ if $x \in A$ and $y \in B$.

Then $(A \dot{\cup} B, \leq)$ is a well-ordered set.

If A and B are ordinal numbers, then the *sum* of A and B is defined as $A + B = \text{ord}(A \dot{\cup} B, \leq)$, where $(A \dot{\cup} B, \leq)$ is the well-ordered set of A8.1A7.

90. Theorem: Let A and B be ordinal numbers. Define $\leq$ on $A \times B$: for $(a, b), (a', b') \in A \times B$, let $(a, b) \leq (a', b')$, provided that either $a \subset a'$ or $a = a'$ and $b \subseteq b'$. Then $(A \times B, \leq)$ is a well-ordered set.

The order on $A \times B$ in A8.18 is called the *lexicographic order*.

If A and B are ordinal numbers, then the *product* BA is defined as $BA = \text{ord}(A \times B, \leq)$ where $\leq$ is the lexicographic order on $A \times B$.

10 Cardinal Numbers

Two sets X and Y are said to be *equipotent* if there is a one-to-one function from X onto Y . Note that equipotence is an equivalence relation on the class of all sets. An equivalence class of this relation is called an *equipotence class*.

If X is a set, then the family of all subsets of X is denoted $\mathcal{P}(X)$ and the family of all functions from X into $\{0, 1\}$ is denoted by 2^X .

91. Theorem: If X is a set, then $\mathcal{P}(X)$ and 2^X are equipotent.

92. Theorem: If $f : X \rightarrow Y$ is a function from X onto Y , then Y is equipotent to a subset of X .

If X is a set, let $\mathcal{E}_X$ denote the equipotence class of X . Note that since X can be well-ordered, $\mathcal{E}_X$ contains at least one ordinal number. The *cardinal number* of X is defined to be the first ordinal number in $\mathcal{E}_X$ (see A5.7) and is denoted $\#X$.

93. Theorem: Let X and Y be sets. Then X is equipotent to a subset of Y if and only if $\#X \subseteq \#Y$. In particular, X and Y are equipotent if and only if $\#X = \#Y$.

94. The Schröder-Bernstein Theorem: If X and Y are sets, and there exist one-to-one functions $f : X \rightarrow Y$ and $g : Y \rightarrow X$, then there exists a one-to-one function $h : X \rightarrow Y$ from X onto Y .

95. Theorem: If X is a set, then $\#X \subset \#\mathcal{P}(X)$.

If X is an infinite set, then $\#X$ is called a *transfinite cardinal number*.

96. Theorem: In the order $\subseteq$ on the class of all cardinal numbers, $\text{ord } \bar{0}$ is the smallest cardinal number, and $\aleph_0$ is the smallest transfinite cardinal number.

97. Theorem: The class of all cardinal numbers is not a set.

We use c to denote $\#\mathbb{R}$.

Observe that $\mathcal{P}(\mathbb{N})$ and $2^{\mathbb{N}}$ are equipotent (A91), and hence $\#\mathcal{P}(\mathbb{N}) = \#2^{\mathbb{N}}$ (A94). Now $\#\mathcal{P}(\mathbb{N}) = \#\mathbb{R} = c$ (A98). In view of A9.9, we see that $\#\mathbb{N}$ is a proper subset of $\#\mathcal{P}(\mathbb{N})$, and we conclude that $\aleph_0$ is a proper subset of c .

The Continuum Hypothesis: If k is a cardinal number and $\aleph_0 \subseteq k \subseteq c$, then either $k = \aleph_0$ or $k = c$; i.e., there is no cardinal number between $\aleph_0$ and c .

The Generalized Continuum Hypothesis: If X is an infinite set and k is a cardinal number such that $\#X \subseteq k \subseteq \#\mathcal{P}(X)$ (see A99), then either $k = \#X$ or $k = \#\mathcal{P}(X)$.

98. Theorem: If $A = \#X$, then $2^A = \#\mathcal{P}(X)$. Moreover, $A \subset 2^A$.

99. Theorem: If A is a transfinite cardinal number, then $A^2 = A$.

100. Corollary: Let A and B be non-empty cardinal numbers, at least one of which is transfinite. Then $AB = A + B = \max\{A, B\}$.

If A is a transfinite cardinal number and $C_A = \{C : C \text{ is a cardinal number and } C \subset A\}$, then the *aleph index* of A is $i(A) = \text{ord}(C_A, \subseteq)$.

101. Theorem: Let A and B be transfinite cardinal numbers. Then:

- (a) $A = B$ if and only if $i(A) = i(B)$;
- (b) $A \subseteq B$ if and only if $i(A) \subseteq i(B)$; and
- (c) $A \subset B$ if and only if $i(A) \subset i(B)$.

102. Theorem: If B is an ordinal number, then there exists a unique transfinite cardinal number A such that $B = i(A)$.

The cardinal number corresponding to an ordinal number B in A9.12 is denoted $\aleph_B$.

103. Theorem: If B is an ordinal number and A is a cardinal number such that $\aleph_B \subseteq A \subseteq \aleph_{B+1}$, then $A = \aleph_A$ or $A = \aleph_{A+1}$.

104. Theorem: If B is an ordinal number and A is a non-empty cardinal number, then $\aleph_{B+1}^A = \aleph_B^A \aleph_{B+1}$.